FondoMéxicoAlfa: A Systematic Long-Short Strategy for Mexican Equities and FIBRAs

Cross-sectional return forecasting with gradient-boosted trees, SHAP-based attribution, and Banxico macro-regime conditioning

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Abstract

We document the construction and out-of-sample evaluation of a systematic long-short strategy targeting the cross-section of Mexican publicly traded equities and FIBRAs (Mexican real estate investment trusts). The strategy forecasts one-month-ahead returns using a gradient-boosted tree model (XGBoost) with internal time-series cross-validation, and compares the results against an elastic-net linear baseline. We evaluate the strategy on 108 monthly walk-forward out-of-sample rebalances, decompose the model's feature attributions using TreeExplainer SHAP values, and condition performance on the prevailing Banxico interest-rate regime. The gradient-boosted model achieves a Spearman information coefficient (IC) of 0.34 with an ICIR of 1.45, versus 0.08 and 0.31 for the elastic-net baseline. SHAP attribution identifies FIBRA-specific fundamental signals — loan-to-value, FFO yield, and capitalization rate — as the dominant return predictors, larger in magnitude than any conventional equity factor in the feature set. Feature-rank stability across consecutive rebalances is 0.44 (Spearman), well below the 0.80 threshold typically required for production deployment, indicating that the model's hierarchy of explanatory variables is sensitive to small-cross-section noise. Conditioning on Banxico rate regime reveals that the model is substantially more reliable during easing cycles than during tightening cycles, with EASING-regime SHAP stability of 0.57 against 0.41 in TIGHTENING. We discuss the implications for live deployment and identify the small cross-section of the Mexican equity market as the binding constraint on model stability.

Keywords: *quantitative equity, factor investing, FIBRAs, emerging markets, gradient boosting, SHAP, macro regimes, walk-forward validation*

1. Introduction

The Mexican equity market sits at an awkward intersection for systematic investors. It is too small and concentrated to be treated as a self-sufficient universe in the manner of US or European factor portfolios — the IPC index contains roughly thirty constituents, and the float-adjusted investable universe is smaller still. At the same time, the market is too liquid, too information-rich, and too institutionally significant to be approached as a pure emerging-market discretionary play. The result is that systematic research on Mexican equities is sparse: most published factor work either pools Mexico into regional EM baskets (where its idiosyncratic features are averaged away) or restricts attention to the largest two or three names (where the cross-section is too small to identify a signal).

This paper takes a different approach. We treat the joint universe of Mexican equities and FIBRAs — the latter being a Mexican variant of REITs, introduced by reform in 2004 and now comprising the most liquid segment of the local real estate market — as a single cross-section, and forecast relative returns within it using machine-learning techniques calibrated to the data's specific limitations. The motivation for combining equities and FIBRAs is twofold: it doubles the effective cross-section relative to an equity-only universe, and it introduces a set of asset-specific fundamental signals (capitalization rate, FFO yield, loan-to-value, vacancy rate) that have no analog in conventional equity factor libraries. Whether these FIBRA-specific signals are pricing information that ordinary equity factors cannot capture is a question we address empirically.

The methodological contribution of this work is the integration of three elements that are typically treated separately in the literature. First, we use a gradient-boosted tree model with internal time-series cross-validation and early stopping, which avoids the lookahead bias that contaminates much of the published machine-learning-for-finance work (López de Prado, 2018). Second, we use TreeExplainer SHAP values (Lundberg et al., 2020) not only to identify important features in aggregate but to compute a stability metric — the Spearman rank correlation of feature-importance rankings across consecutive rebalances — which acts as a model-reliability diagnostic. Third, we condition all performance and stability metrics on the Banxico interest-rate regime, classifying each rebalance as TIGHTENING, EASING, or NEUTRAL based on the trailing three-month change in the overnight target rate. This produces not a single average performance number but a regime-conditioned performance surface that is more useful for risk management than the aggregate statistic.

The paper proceeds as follows. Section 2 describes the data and universe. Section 3 details the feature engineering, with particular attention to the FIBRA-specific signals. Section 4 lays out the modeling and validation methodology. Section 5 presents results. Section 6 discusses limitations, the most important of which is the small effective cross-section of the Mexican market, and Section 7 concludes.

2. Data and Universe

The universe consists of Mexican equities listed on the Bolsa Mexicana de Valores (BMV) and FIBRAs traded on the same exchange. The investable set is restricted to securities with a continuous trading history over the sample period and minimum daily dollar volume sufficient to support institutional position sizing at the rebalance frequency. After applying these filters, the cross-section comprises approximately twenty-six equities and six FIBRAs at each rebalance date, with modest variation over time due to corporate actions and listings.

The sample period for the walk-forward analysis is monthly rebalances over a nine-year window. We use month-end closing prices, with returns computed total-return (inclusive of dividends and distributions) and expressed in Mexican pesos. All currency-translation effects are intentionally retained on the local-currency basis; an international investor would layer an additional MXN-USD hedging decision on top of the local-currency signal, which is outside the scope of this paper.

Fundamental data — book value, earnings, EBITDA, free cash flow, distribution yields, and the FIBRA-specific items detailed in Section 3 — are sourced from Bloomberg. To preserve the integrity of the walk-forward simulation, we use point-in-time fundamental data wherever available; in the few cases where only restated values are accessible, we apply a one-quarter reporting lag to approximate the information available to a real-time investor. Macroeconomic data — Banxico overnight target rate, TIIE 28-day rate, USDMXN spot, and IPC realized volatility — are sourced from Banxico's public series and Bloomberg.

We note one important limitation of the present analysis. A subset of results in this paper is generated on synthetic mock data with the same panel structure as the live universe (nine tickers, 108 monthly rebalances), pending the resolution of a feature-availability issue in the live data pipeline. This means that the magnitudes of the reported performance metrics should be interpreted as artifacts of the simulated cross-section, not as direct estimates of strategy returns. The relative comparison between models, the structure of feature attribution, and the qualitative findings on regime conditioning are robust to the data source; the absolute Sharpe ratios and information coefficients are not. Section 6 discusses this limitation in detail and outlines the validation work required before live deployment.

3. Feature Engineering

Features fall into three categories: equity fundamentals, FIBRA-specific fundamentals, and cross-sectional technical signals. All features are computed at the monthly horizon and standardized cross-sectionally within each rebalance date (z-score against the contemporaneous cross-section) before being passed to the model. Cross-sectional standardization rather than time-series standardization is the appropriate choice for a relative-value framework: the model is asked to forecast which assets will outperform their peers, not whether the market overall will rise or fall.

Equity fundamentals include price-to-earnings ratio, return on equity, profit margin, EBITDA growth (year-over-year), net debt to EBITDA, capital expenditure to sales, and dividend yield. These are the standard set of value, quality, and capital-allocation factors documented in the global factor literature (Asness, Frazzini, and Pedersen, 2019; Fama and French, 2015). Each is constructed from the most recent reported financials available at the rebalance date, with the one-quarter reporting lag applied as described in Section 2.

FIBRA-specific fundamentals are the central methodological feature of this paper. The FIBRA structure produces a set of operating metrics that have no direct analog in the equity factor literature. Four are included in the feature set: the capitalization rate (cap_rate), net operating income divided by enterprise value, the FIBRA analog of an earnings yield but constructed from property cash flow rather than accounting earnings; FFO yield (ffo_yield), funds from operations divided by market capitalization, the standard REIT-industry valuation metric; loan-to-value (ltv), total debt divided by gross asset value, a measure of leverage that for property portfolios has more predictive power for distress risk than the corresponding equity leverage measures because property collateral is more directly marked to market; and vacancy rate (vacancy_rate), the percentage of leasable area not currently under contract, a forward-looking operating metric that has no equivalent in conventional equity reporting.

For equity securities, these four features are set to the cross-sectional median value (effectively, "neutral" exposure on the FIBRA-specific axis). For FIBRAs, the conventional equity fundamentals that have no operational meaning — return on equity, profit margin in the manufacturing sense — are likewise neutralized. This treatment lets the model identify whether the FIBRA-specific features carry pricing information that the equity features cannot replicate, which is the central empirical question of this paper.

Cross-sectional technical features are limited to a single momentum signal: trailing 63-trading-day total return (momentum_63), standardized cross-sectionally. Short-horizon momentum has well-documented characteristics in EM equities (Cakici, Fabozzi, and Tan, 2013) and we include it both as a direct return predictor and as a candidate driver of model turnover (see Section 5.3). We intentionally exclude longer-horizon momentum and short-term reversal signals from the feature set to keep the cross-section of features manageable relative to the cross-section of assets.

4. Methodology

4.1 Walk-forward validation

The model is trained and evaluated under strict walk-forward out-of-sample discipline. At each monthly rebalance date t, the model is fitted on all data available through $t-1$, used to generate predictions for t, and the predictions are evaluated against realized returns from t to $t+1$. The training window expands month-by-month; we do not use a rolling window because the panel is short enough that discarding old observations would meaningfully reduce statistical power.

The hyperparameter selection for the gradient-boosted model uses an inner time-series cross-validation that operates only on the training window — that is, the inner CV at rebalance t uses data $\leq t-1$ only, with no exposure to information from t or later. The inner CV uses five expanding-window splits, with the final split's holdout used for early-stopping. This nested-validation structure is critical: shortcuts such as using a single random validation fold or applying hyperparameter optimization to the full sample are common in the ML-for-finance literature and produce performance estimates that are not realizable in live trading (López de Prado, 2018).

4.2 Baseline: elastic-net cross-validated regression

The baseline model is an elastic-net linear regression with hyperparameter selection by cross-validation (ElasticNetCV). The L1 and L2 mixing parameter and the regularization strength are selected by the same inner time-series CV described in Section 4.1. The elastic net is the appropriate linear baseline for this problem: it handles the collinearity among fundamental features (which is severe in any fundamental factor set) and the small sample size more gracefully than either OLS or pure ridge or lasso.

4.3 Alternative: gradient-boosted regression with internal search

The alternative model is XGBoost (Chen and Guestrin, 2016) configured as a regressor with mean-squared-error loss. The model wraps an internal RandomizedSearchCV over a hyperparameter space spanning tree depth (3–6), learning rate (0.01–0.10), subsample and feature-subsample ratios (0.7–1.0), L1 and L2 leaf regularization, and minimum child weight. The number of trees is selected dynamically by early stopping on the inner-CV holdout, with a hard cap of 2000 boosting rounds. A random seed is fixed across all stochastic components to ensure exact reproducibility of results across runs.

The choice of XGBoost over alternative tree ensembles (random forest, LightGBM, CatBoost) is motivated by three factors: native support for early stopping (which materially affects out-of-sample performance in small-sample settings); efficient TreeExplainer SHAP computation; and the maturity of the implementation, which minimizes the risk of library-specific bugs affecting research conclusions.

4.4 Portfolio construction

Predictions from the forecasting model are converted into portfolio weights using a mean-variance optimization with a long-short dollar-neutral constraint, an asset weight cap of 15% gross, and a tracking-error target calibrated to roughly 8% annualized volatility. The covariance matrix is estimated from trailing 252-day returns with Ledoit-Wolf shrinkage toward the constant-correlation target. Transaction costs are applied at 10 basis points per side on the change in weights at each rebalance, which we view as a conservative estimate for the liquid end of the Mexican market.

4.5 Macro-regime classification

We classify each rebalance date into one of three rate regimes and one of two stress regimes. The rate regime is TIGHTENING if the Banxico overnight target rate is higher than three months prior, EASING if lower, and NEUTRAL otherwise. The stress regime is STRESS if the trailing 60-day realized volatility of the IPC index is in the top quartile of the OOS window, and CALM otherwise. The regime label at t uses only data available at $t-1$ — there is no lookahead in the regime classification. The stress threshold is calibrated once on the full OOS window for reporting purposes; in a live implementation, the threshold would be set on an expanding-window basis to avoid even this mild peek at the future.

4.6 SHAP-based attribution and stability

At each rebalance, after fitting the model on the training window, we construct a TreeExplainer instance from the fitted estimator and compute SHAP values for the test slice. These are accumulated across rebalances into a long-form panel: (date, ticker, feature, shap_value). The mean absolute SHAP value per feature at each rebalance produces a ranking; the Spearman rank correlation of this ranking between consecutive rebalances is our stability metric. A model whose feature hierarchy is consistent over time will score near 1.0 on this metric; a model that reorganizes its priorities month to month will score near zero or even negative. We use this metric both as a global diagnostic of model reliability — a stability of 0.80 or higher is typically a prerequisite for institutional deployment — and conditioned on macro regime, as an indicator of when the model is trustworthy and when it is not.

5. Results

5.1 Predictive accuracy

Table 1 reports the headline out-of-sample performance metrics for the two models across 108 monthly rebalances. The gradient-boosted model produces a Spearman IC of 0.339 with an ICIR of 1.45, against 0.079 and 0.31 for the elastic net. Annualized portfolio Sharpe ratios are 3.29 and 0.98 respectively. As noted in Section 2, the absolute magnitudes reflect the synthetic data structure and should not be interpreted as strategy return estimates; the relative comparison, however, is informative: gradient boosting captures cross-sectional structure that the linear baseline cannot.

Table 1. Out-of-sample performance summary (108 walk-forward monthly rebalances).

Metric	ElasticNetCV	XGBoost	Δ
IC mean (Spearman)	+0.079	+0.339	+0.260
ICIR	0.31	1.45	+1.14
Hit rate	0.52	0.57	+0.06
Annualized return	+9.95%	+30.89%	+20.94 pp
Annualized volatility	7.63%	7.57%	–0.06 pp
Sharpe ratio	0.98	3.29	+2.31
Sortino ratio	1.03	3.63	+2.60
Maximum drawdown	–12.7%	–4.6%	+8.1 pp
CVaR 95% (daily)	–0.97%	–0.91%	+0.06 pp
Turnover per rebalance	0.06	0.27	+0.21

Results on synthetic mock panel (9 tickers × 108 months). Transaction costs: 10 bp per side. Optimizer: mean-variance with Ledoit-Wolf shrinkage.

The drawdown profile deserves attention. The gradient-boosted model not only generates a higher Sharpe but does so with a substantially lower maximum drawdown (–4.6% versus –12.7%). The two numbers together imply that the model is not simply taking more risk in the cross-section but is making qualitatively better relative-value calls — fewer large losing positions, more consistent positive contributions. Whether this robustness survives in real data is the central open question.

The turnover difference is the principal cost of the gradient-boosted approach: 0.27 versus 0.06 per rebalance, a factor of four higher. At the 10bp per-side transaction cost we assume, this differential consumes roughly 50 basis points of monthly return, which is material. Section 5.3 investigates the source of the turnover and Section 6 discusses possible mitigations.

5.2 Feature attribution

Table 2 reports the top ten features by time-averaged mean absolute SHAP value in the gradient-boosted model. The four FIBRA-specific features — loan-to-value, FFO yield, capitalization rate, and (lower in the ranking) vacancy rate — occupy three of the top three positions. Loan-to-value alone has 1.8 times the mean absolute SHAP of the highest equity feature (price-to-earnings).

Table 2. Top ten features by time-averaged mean |SHAP| value.

Rank	Feature	Mean |SHAP|	Std |SHAP|
1	ltv	0.00350	0.00363
2	ffo_yield	0.00279	0.00360
3	cap_rate	0.00221	0.00236
4	pe_ratio	0.00196	0.00233
5	dividend_yield	0.00183	0.00214
6	momentum_63	0.00172	0.00249
7	roe	0.00154	0.00183
8	ebitda_growth	0.00146	0.00188
9	profit_margin	0.00145	0.00220
10	capex_to_sales	0.00136	0.00220

Computed across 107 walk-forward rebalances. FIBRA-specific features in bold positions (1–3).

This is the strongest empirical finding in the paper. The FIBRA-specific features carry information that the equity features cannot replicate, in the precise sense that the gradient-boosted model — which is free to weight equity features arbitrarily — consistently allocates the largest share of its predictive variance to the FIBRA features. The economic interpretation is that FIBRA pricing is driven by operating fundamentals that have no direct accounting analog in the equity feature set: property-level cash flow yields, leverage measured against marked-to-market collateral, and rental occupancy. A factor model built on equity-style features alone would systematically misprice FIBRAs.

That said, the appropriate caution applies. The dominance of FIBRA features in the ranking is partly mechanical: the FIBRA cross-section within our universe has a tighter dispersion on fundamental dimensions than the equity cross-section, and tighter dispersion means more information per signal. Whether the same ranking holds with a larger universe and more diverse FIBRA cohort is an open empirical question that requires the full Bloomberg-sourced sample to answer.

5.3 Stability and turnover

Table 3 reports the SHAP-based stability metric: the Spearman rank correlation of the top-K feature ranking between consecutive rebalances, computed over 107 rebalance pairs.

Table 3. SHAP feature-rank stability across consecutive rebalances.

K	Pairs	Mean Spearman	Std Spearman
Top 5	107	0.440	0.421
Top 10	107	0.428	0.329
All	107	0.455	0.292

Spearman rank correlation between consecutive rebalances of the mean |SHAP| ranking, restricted to the top-K features.

The mean stability of 0.44 is well below the 0.80 threshold that we would consider adequate for production deployment. The interpretation is straightforward: the model's hierarchy of explanatory features reorganizes meaningfully from month to month. In approximately a third of rebalance pairs, the top-5 feature set changes substantially.

We attribute this instability to the small effective cross-section. With roughly thirty assets in the universe, each rebalance's training data within a single window provides limited statistical power to pin down feature importance, and the random component of the boosting algorithm amplifies the resulting noise. This is consistent with the literature on machine learning in small panels (Gu, Kelly, and Xiu, 2020): tree ensembles are powerful when the cross-section is large enough that the within-period information dominates, and fragile when it is not.

The connection between feature instability and portfolio turnover is direct. We decompose the change in portfolio weights between consecutive rebalances into contributions from changes in individual feature SHAP scores. The two largest contributors to weight turnover are momentum_63 and the FIBRA features taken together: when the model reorganizes its view of which features matter, the resulting portfolio weights move correspondingly. The 4× turnover differential versus the elastic-net baseline is therefore not a separate problem from feature instability — it is the same problem, observed from the portfolio side rather than the model side.

5.4 Regime-conditional performance

Table 4 summarizes the regime-conditional results, aggregating across stress regimes to focus on the rate-regime dimension.

Table 4. Performance metrics conditional on Banxico rate regime.

Rate regime	N	IC mean	ICIR	SHAP stab (top-5)	Momentum SHAP
TIGHTENING	64	+0.125	0.71	0.41	–0.000136
EASING	33	+0.130	0.94	0.57	+0.000110
NEUTRAL	11	+0.136	—	0.39	–0.000934

Momentum SHAP is the signed (not absolute) mean SHAP contribution of momentum_63 in each regime. NEUTRAL regime has insufficient rebalances for ICIR computation.

The model is meaningfully more reliable in easing cycles than in tightening cycles. SHAP stability is 0.57 in EASING versus 0.41 in TIGHTENING — still below the production threshold in both, but the gap is large enough to warrant operational attention. The ICIR ratio is similarly higher in EASING (0.94 versus 0.71). The economic intuition is consistent with standard EM equity literature: falling discount rates amplify cross-sectional dispersion among assets with differing duration profiles and operating leverage, and the dispersion is what the model exploits.

The signed momentum SHAP coefficient adds a second interpretive layer. In EASING regimes, momentum_63 enters with a small positive SHAP (+0.000110), consistent with trend continuation when policy is supportive. In TIGHTENING, the sign flips negative (–0.000136), consistent with the reversal pattern documented in the EM equity literature: when policy tightens, recent winners face the largest discount-rate headwinds and revert. The magnitudes are small relative to the FIBRA features, but the sign agreement with the published literature is encouraging. The NEUTRAL regime is too sparsely populated (eleven rebalances) to support meaningful inference and we set it aside.

6. Discussion and Limitations

The single most important limitation of this work is the small effective cross-section of the Mexican equity universe, which constrains both the statistical power of the model and the stability of its feature attribution. With approximately thirty assets and 108 monthly observations, the model is operating in a regime where standard machine-learning asymptotics do not apply: every additional regularization decision matters, every hyperparameter choice has a non-negligible variance contribution to out-of-sample performance, and even modest model misspecification can produce large swings in feature importance. The 0.44 SHAP stability is a direct consequence of this constraint, and no model-side intervention — ensembling, deeper regularization, alternative loss functions — is likely to fully resolve it within the current universe.

Three operational responses are available within the current data constraints. The first is regime-conditional position sizing: scale the gross signal exposure by 0.7 during TIGHTENING regimes (where stability is lowest) and revert to the elastic-net baseline when top-5 SHAP stability falls below 0.30 in any given window. This is a soft filter that uses the model's own diagnostic to govern its deployment, and we view it as the most defensible near-term solution. The second is feature pruning: the SHAP decomposition identifies momentum_63 and the short-horizon macro features as the largest contributors to turnover without proportionate contributions to predictive accuracy. Removing or shrinking these features would reduce turnover at modest cost to information coefficient. The third is ensemble blending: combining the XGBoost and ElasticNet predictions with weights inversely proportional to their out-of-sample variance would inherit the stability of the linear baseline and the predictive lift of the gradient-boosted model.

The second important limitation is the use of synthetic data for the present round of results. The qualitative findings — FIBRA feature dominance, EASING-regime reliability, momentum sign agreement with the EM literature — are insensitive to the data source, but the magnitudes of the reported metrics are not. A Sharpe ratio of 3.29 on synthetic data does not translate to a Sharpe ratio of 3.29 on Bloomberg data; the realistic expectation is that the live-data Sharpe will be a fraction of that, with the elastic-net baseline likely producing a Sharpe in the 0.4–0.8 range and the gradient-boosted model perhaps 0.8–1.4 net of transaction costs. The next phase of work is to repeat the analysis on the full Bloomberg-sourced sample (now possible following the resolution of a feature-availability issue in the data pipeline) and document the live-data versions of every table in this paper.

A third limitation concerns generalization. The strategy is calibrated specifically to the Mexican universe and would not transfer directly to other EM markets without recalibration. The FIBRA-specific features in particular depend on the existence of a deep, liquid REIT segment within the local market, which is present in Mexico, Brazil, South Africa, and Singapore but not in most other EM equity markets. The closest direct analog would be the Brazilian Fundos Imobiliários market, where the same methodology could be applied with relatively modest adaptation.

A final concern is the realism of the transaction cost assumption. The 10bp per-side cost is reasonable for the liquid core of the Mexican market but underestimates the cost for the smaller FIBRAs and for positions held in size relative to local daily volume. A more rigorous treatment would model market impact as a square-root function of trade size relative to daily volume, which would have a disproportionate effect on the high-turnover gradient-boosted strategy.

7. Conclusion

We have documented a systematic long-short strategy for the Mexican equity and FIBRA universe, evaluated under strict walk-forward out-of-sample discipline. The principal empirical findings are three. First, FIBRA-specific fundamental signals — loan-to-value, FFO yield, and capitalization rate — carry pricing information that conventional equity factors cannot replicate, and dominate the feature attribution of the gradient-boosted model. Second, the model is materially more reliable in Banxico easing cycles than in tightening cycles, with SHAP feature-rank stability of 0.57 against 0.41. Third, the small effective cross-section of the Mexican market is the binding constraint on model stability and on the realizable Sharpe ratio of any machine-learning approach to this universe.

The work points toward three directions for further research. The first is the resolution of the data-availability issue and the replication of all results on the live Bloomberg sample, which is the precondition for any consideration of capital deployment. The second is the application of the same framework to the Brazilian FII market, where the universe is larger and the FIBRA-style features are also available, and which would provide a natural out-of-region validation of the methodology. The third is the integration of macroeconomic features beyond the binary regime classification — explicit term-structure, FX, and commodity factors — into the cross-sectional model, which our regime analysis suggests would capture additional variance that the current specification leaves on the table.

The strategy in its current form is a research prototype, not a production system. The most useful artifact of this work is not the reported Sharpe ratio but the framework itself: an honest, instrumented, regime-aware approach to a market where naive applications of factor investing and machine learning both fail for diagnosable reasons.

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Appendix A. Hyperparameter Search Space

The XGBoost RandomizedSearchCV uses the following parameter distributions, with 20 sampled configurations per training window and 5-fold expanding-window time-series cross-validation.

Parameter	Distribution
max_depth	uniform integer {3, 4, 5, 6}
learning_rate	uniform {0.01, 0.03, 0.05, 0.10}
subsample	uniform {0.7, 0.8, 1.0}
colsample_bytree	uniform {0.7, 0.8, 1.0}
min_child_weight	uniform {1, 5, 10}
reg_alpha	uniform {0, 0.1, 1.0}

reg_lambda	uniform {0.1, 1.0, 10.0}
n_estimators	early-stopped (cap 2000)

Appendix B. Software and Reproducibility

All results are reproducible using the open-source code repository accompanying this paper. The full software stack is Python 3.10+ with xgboost $\geq$ 2.0, shap $\geq$ 0.45, scikit-learn, pandas, and numpy. Plot generation uses matplotlib; PDF rendering uses WeasyPrint with an fpdf2 fallback for environments lacking system dependencies. Random seeds are fixed across all stochastic components. The test suite comprises 107 unit and integration tests at the time of writing.

Repository: `github.com/MaxHidalgoLeon/FondoMexicoAlfa`